Association Analysis: Basic Concepts

Association Rule Mining

- Given a set of transactions, find rules that will predict the occurrence of an item based on the occurrences of other items in the transaction

Market-Basket transactions

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Association Rules

$\{Diaper\} \rightarrow \{Beer\}$,
$\{Milk, Bread\} \rightarrow \{Eggs,Coke\}$,
$\{Beer, Bread\} \rightarrow \{Milk\}$,

Definition: Frequent Itemset

- **Itemset**
  - A collection of one or more items
    - Example: {Milk, Bread, Diaper}
  - k-itemset
    - An itemset that contains k items
- **Support count (σ)**
  - Frequency of occurrence of an itemset
  - E.g. σ({Milk, Bread,Diaper}) = 2
- **Support**
  - Fraction of transactions that contain an itemset
  - E.g. s({Milk, Bread, Diaper}) = 2/5
- **Frequent Itemset**
  - An itemset whose support is greater than or equal to a *minsup* threshold

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Definition: Association Rule

- **Association Rule**
  - An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets
  - Example:
    $\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\}$

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
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- **Rule Evaluation Metrics**
  - Support (s)
    - Fraction of transactions that contain both X and Y
  - Confidence (c)
    - Measures how often items in Y appear in transactions that contain X

Example:

$$\{\text{Milk}, \text{Diaper}\} \Rightarrow \{\text{Beer}\}$$

$$s = \frac{\sigma(\text{Milk}, \text{Diaper}, \text{Beer})}{|\text{T}|} = \frac{2}{5} = 0.4$$

$$c = \frac{\sigma(\text{Milk}, \text{Diaper}, \text{Beer})}{\sigma(\text{Milk}, \text{Diaper})} = \frac{2}{3} = 0.67$$

Association Rule Mining Task

- Given a set of transactions T, the goal of association rule mining is to find all rules having
  - support ≥ *minsup* threshold
  - confidence ≥ *minconf* threshold

- Brute-force approach:
  - List all possible association rules
  - Compute the support and confidence for each rule
  - Prune rules that fail the *minsup* and *minconf* thresholds

  $\Rightarrow$ Computationally prohibitive!

Reducing Number of Candidates

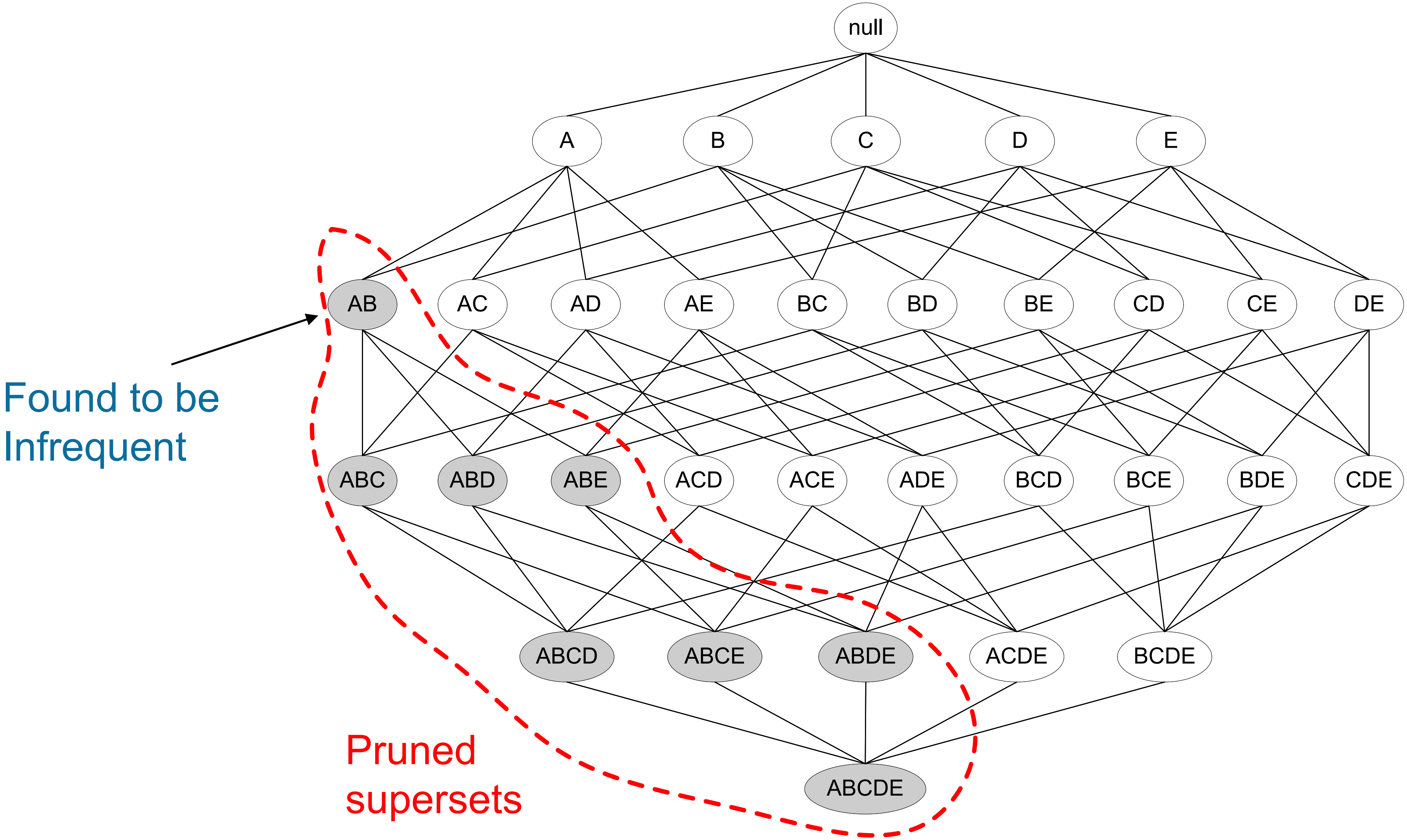
- Apriori principle:
  - If an itemset is frequent, then all of its subsets must also be frequent

- Apriori principle holds due to the following property of the support measure:

$$\forall X, Y : (X \subseteq Y) \Rightarrow s(X) \geq s(Y)$$

  - Support of an itemset never exceeds the support of its subsets
  - This is known as the anti-monotone property of support

Illustrating Apriori Principle



$Sup_{min} = 2$

Database TDB

Tid	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

1[st] scan

C_1

Itemset	sup
{A}	2
{B}	3
{C}	3
{D}	1
{E}	3

L_1

Itemset	sup
{A}	2
{B}	3
{C}	3
{E}	3

C_2

Itemset
{A, B}
{A, C}
{A, E}
{B, C}
{B, E}
{C, E}

2[nd] scan

C_2

Itemset	sup
{A, B}	1
{A, C}	2
{A, E}	1
{B, C}	2
{B, E}	3
{C, E}	2

L_2

Itemset	sup
{A, C}	2
{B, C}	2
{B, E}	3
{C, E}	2

C_3

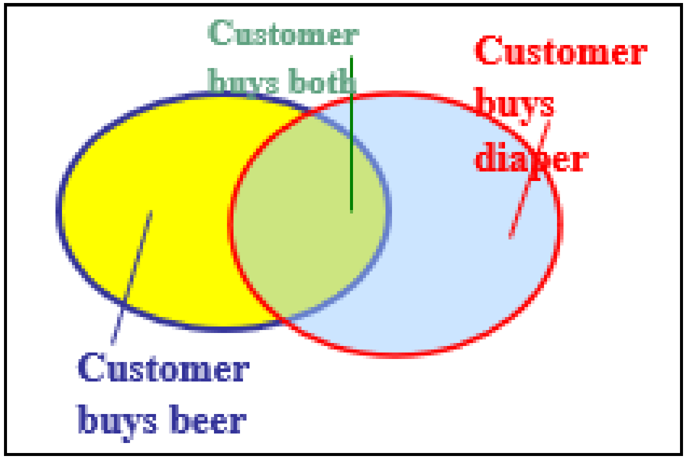
Itemset
{B, C, E}

3[rd] scan

L_3

Itemset	sup
{B, C, E}	2

Tid	Items bought
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diaper
30	Beer, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk



- Find all the rules *X* → *Y* with minimum support and confidence
  - support, *s*, probability that a transaction contains X ∪ Y
  - confidence, *c*, conditional probability that a transaction having X also contains *Y*

Let minsup = 50%, minconf = 50%

Freq. Pat.: Beer:3, Nuts:3, Diaper:4, Eggs:3, {Beer, Diaper}:3

- Association rules: (many more!)
  - *Beer* → *Diaper* (60%, 100%)
  - *Diaper* → *Beer* (60%, 75%)

Common applications of association analysis

1. **Market basket analysis**

   Retailers analyse which products are commonly purchased together.

   Example: Customers who buy **bread and butter** may also frequently buy **milk**.

   This can help with product placement, promotions and cross-selling.

2. **Recommendation systems**

   Association rules can identify items that are frequently selected by similar users.

   Example: "Customers who purchased this laptop also purchased a wireless mouse."

3. **E-commerce and online shopping**

   Online retailers can use associations to create product bundles and suggest complementary products during checkout.

4. **Healthcare**

   Association analysis can identify relationships among diseases, symptoms, medications and patient characteristics.

   Example: Discovering that particular combinations of symptoms frequently occur with a particular condition.

5. **Cybersecurity**

   It can identify combinations of events or behaviours that frequently occur together during suspicious activities.

   Example: Multiple failed logins followed by an unusual successful login and large data transfer may form a suspicious behavioural pattern.

Common applications of association analysis

6. **Banking and fraud detection**

   Financial organisations can discover patterns of transactions associated with fraudulent behaviour.

7. **Web usage mining**

   Websites can analyse pages that users commonly visit together.

   Example: Users who visit a university's course page may frequently visit the fees and scholarship pages next.

8. **Telecommunications**

   Telecom companies can analyse relationships between customer behaviours, service usage and subscription choices to support customer retention and service bundling.

9. **Education analytics**

   Universities can identify associations between student engagement, assessment performance, attendance and academic outcomes.

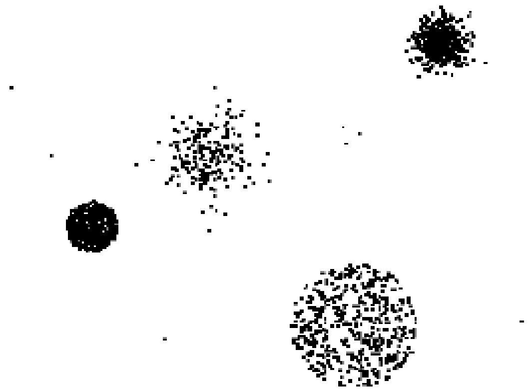
10. **Social media analysis**

    Association analysis can identify topics, hashtags, interests or behaviours that frequently appear together.

Anomaly Detection

Anomaly/Outlier Detection

- What are anomalies/outliers?
  - The set of data points that are considerably different than the remainder of the data



- Natural implication is that anomalies are relatively rare
  - One in a thousand occurs often if you have lots of data
  - Context is important, e.g., freezing temps in July
- Can be important or a nuisance
  - Unusually high blood pressure
  - 200 pound, 2 year old

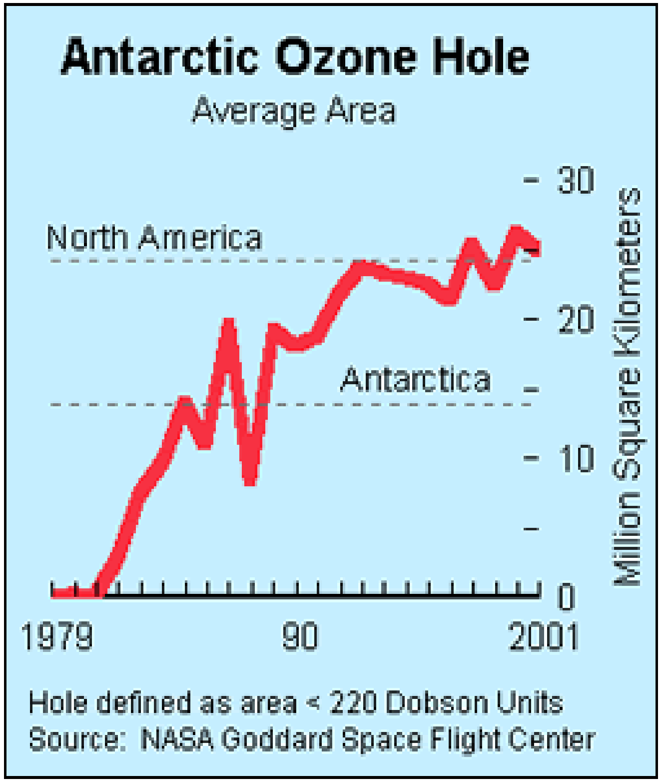
Importance of Anomaly Detection

Ozone Depletion History

- In 1985 three researchers (Farman, Gardinar and Shanklin) were puzzled by data gathered by the British Antarctic Survey showing that ozone levels for Antarctica had dropped 10% below normal levels

- Why did the Nimbus 7 satellite, which had instruments aboard for recording ozone levels, not record similarly low ozone concentrations?

- The ozone concentrations recorded by the satellite were so low they were being treated as outliers by a computer program and discarded!



Source:
http://www.epa.gov/ozone/science/hole/size.html

Causes of Anomalies

- Data from different classes
  – Measuring the weights of oranges, but a few grapefruit are mixed in

- Natural variation
  – Unusually tall people

- Data errors
  – 200 pound 2 year old

Distinction Between Noise and Anomalies

- Noise doesn’t necessarily produce unusual values or objects
- Noise is not interesting
- Noise and anomalies are related but distinct concepts

Common applications of Outlier analysis/Anomaly Detection

1. **Cybersecurity and Intrusion Detection**
   - Detecting unusual network traffic
   - Identifying unauthorised access attempts
   - Detecting compromised user accounts
   - Identifying malware or abnormal system behaviour
   - Detecting unusual login locations or times

   Example: A user normally logs in from Sydney during business hours. A login suddenly occurs from another country at 3:00 am with unusually high data downloads. This could be flagged as an anomaly.
2. **Banking and Financial Fraud**
   - Credit/debit card fraud detection
   - Suspicious banking transactions
   - Money laundering detection
   - Insurance claim fraud

   Example: A customer normally spends \$50–\$200 per transaction but suddenly makes several transactions worth thousands of dollars. The transactions may be treated as anomalies.
3. **Healthcare**
   - Detecting abnormal patient vital signs
   - Identifying unusual laboratory results
   - Detecting abnormalities in medical images
   - Monitoring patients using wearable devices

   Example: An unexpected change in heart rate, blood pressure or oxygen level can trigger an alert.

Common applications of Outlier analysis/Anomaly Detection

4. **Manufacturing and Predictive Maintenance**
    - Detecting faulty equipment
    - Identifying abnormal sensor readings
    - Predicting machine failure
    - Quality control and defect detection

    Example: A machine normally operates between 60–70°C. A sudden increase to 100°C may indicate an impending failure.
5. **IoT and Smart Systems**
    - Detecting malfunctioning IoT devices
    - Identifying abnormal sensor behaviour
    - Smart home security
    - Smart agriculture monitoring

    Example: A soil moisture sensor producing values dramatically different from nearby sensors could indicate a faulty device.
6. **Telecommunications**
    - Network fault detection
    - Abnormal bandwidth consumption
    - Detecting fraudulent mobile usage
    - Identifying service degradation
7. **E-commerce**
    - Detecting fraudulent purchases
    - Identifying fake reviews
    - Unusual customer behaviour
    - Detecting bot activity

Common applications of Outlier analysis/Anomaly Detection

8. **Transportation**
    - Detecting unusual driving behaviour
    - Vehicle fault detection
    - Traffic anomaly detection
    - Identifying abnormal GPS patterns
9. **Social Media**
    - Spam detection
    - Fake account identification
    - Bot detection
    - Detecting unusual activity or misinformation campaigns
10. **Data Quality and Data Science**

- Identifying incorrect data entries
- Detecting measurement errors
- Finding corrupted records
- Identifying unusual observations before building machine-learning models

A simple way to understand it is:

Normal behaviour → expected pattern

Outlier/anomaly → significant deviation from that pattern